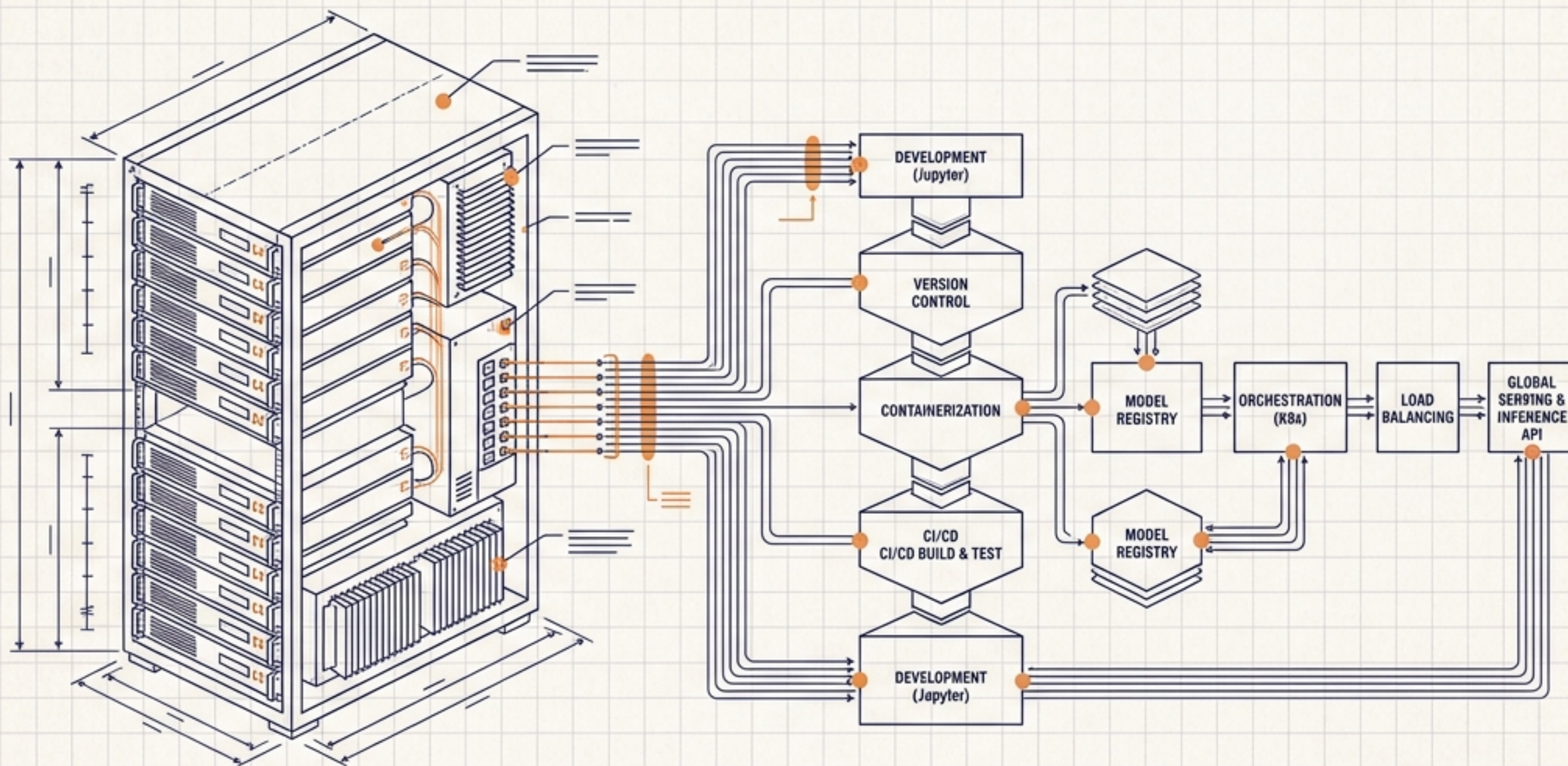


BUILDING INFERENCE SYSTEMS AT SCALE

From Jupyter Notebooks to Serving Millions of Users



THE REAL CHALLENGE BEGINS AFTER TRAINING

THE LABORATORY



Data scientists focus on accuracy and convergence in isolated local environments.

THE REAL WORLD



AI engineers focus on speed, cost, and reliability under the pressure of live traffic.

Moving a model from a local test to a live application is a massive engineering hurdle. The ultimate goal is building robust systems that handle unpredictable inputs and serve millions without crashing.

THE APPLICATION PHASE OF AI

Definition

Inference is the process of applying a trained AI model to real-world input data to generate predictions. While training acts as 'education', inference is the 'application' where models actually deliver business value.



Conversational AI: Turning a user's text prompt into a generated natural language response.



Fraud Detection: Instantly scoring a financial transaction with a probability of being fraudulent.

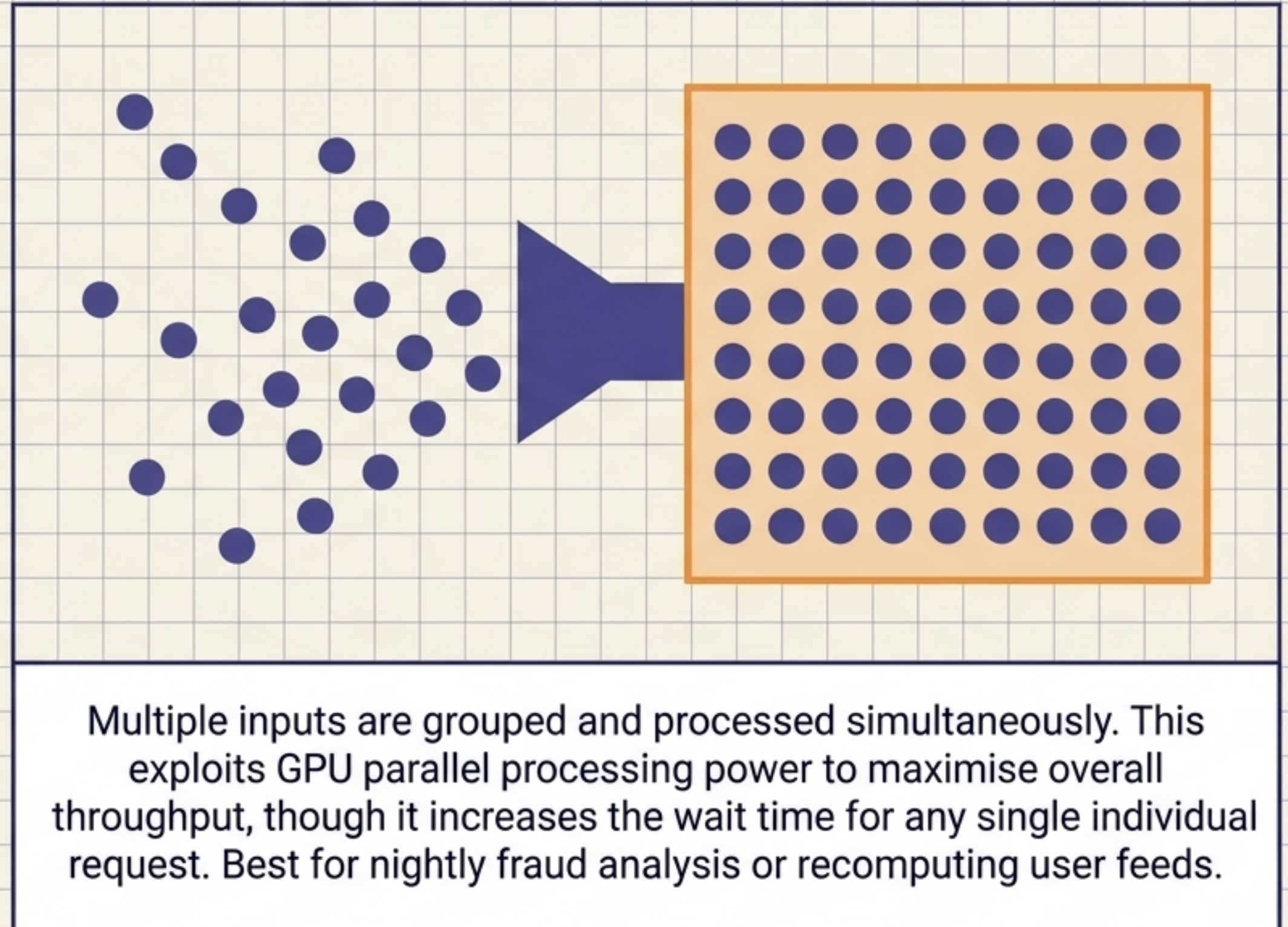
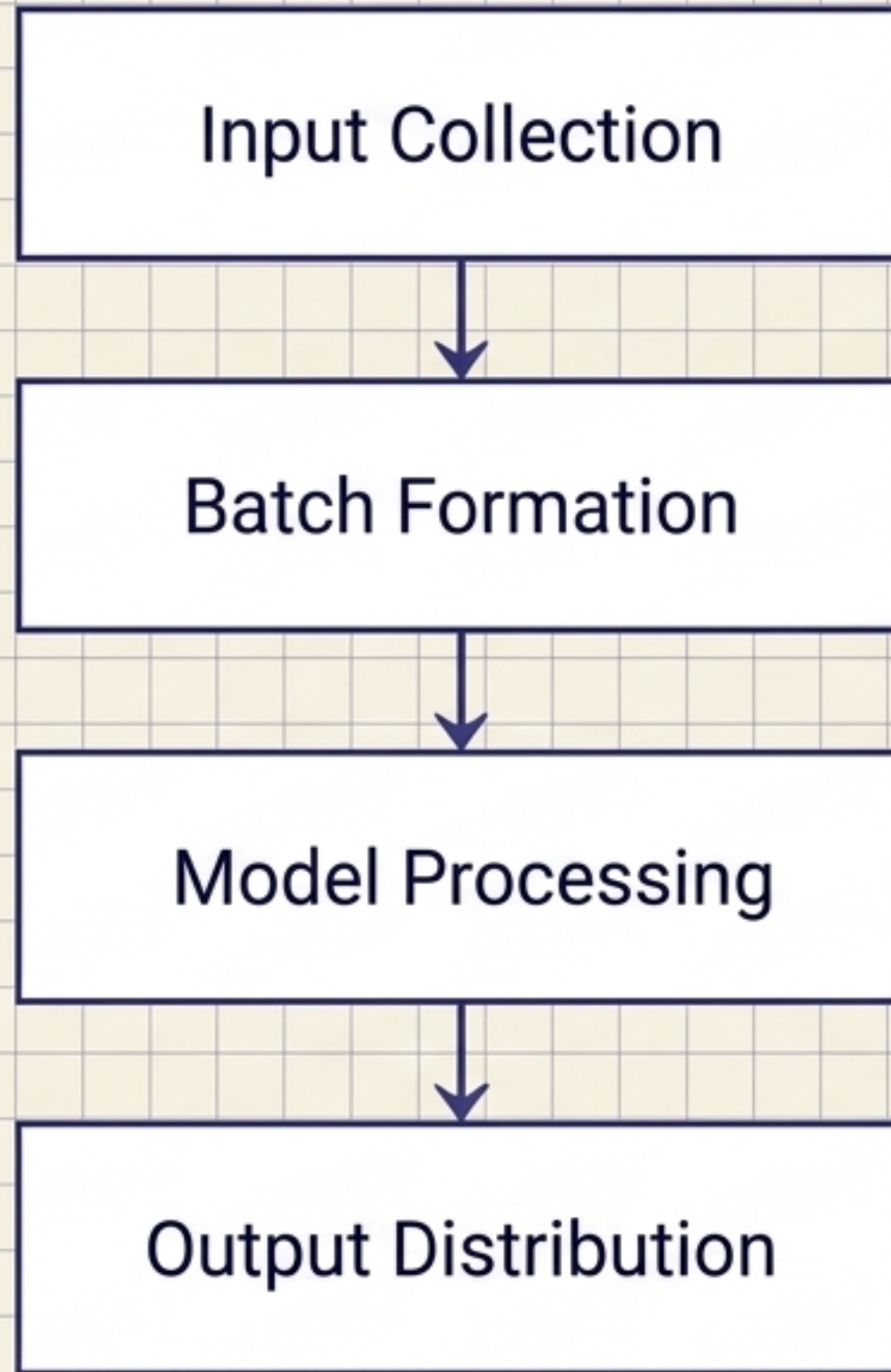


Medical Imaging: Analysing a radiology scan to classify diseases and provide confidence scores.

TRAINING VS. INFERENCE REQUIREMENTS

 TRAINING	 INFERENCE
 Learns patterns from historical data	Applies those patterns to new inputs
 Hours or days on massive GPU clusters	Milliseconds per individual request
 Highly tolerant of slow processing speeds	Requires strict, rapid response times
 A failed job is simply restarted	Results in an immediate user error
 Aims for convergence and accuracy	Optimises for throughput and cost

PARADIGM 1: BATCH INFERENCE



PARADIGM 2: REAL-TIME INFERENCE



The system responds to each individual request as it arrives. It is highly sensitive to network delays, cold-start hardware times, and sudden spikes in traffic.

Fraud Detection

Under 100ms. Transactions must be approved before checkout completes.

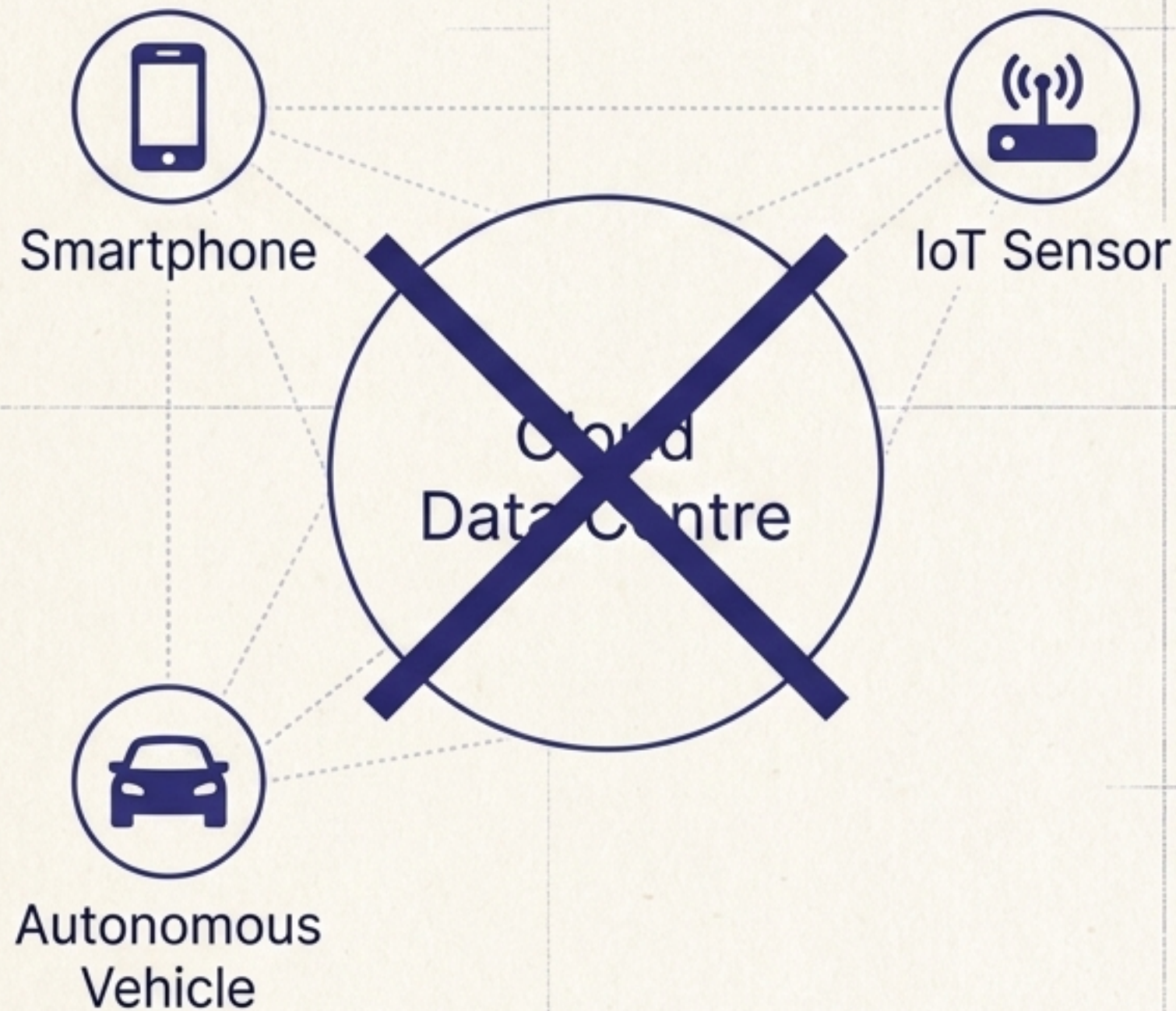
Voice Assistants

Under 300ms. Required to avoid breaking the conversational flow.

Live Translation

Under 500ms. Natural conversation requires near-instantaneous processing.

PARADIGM 3: EDGE INFERERENCE

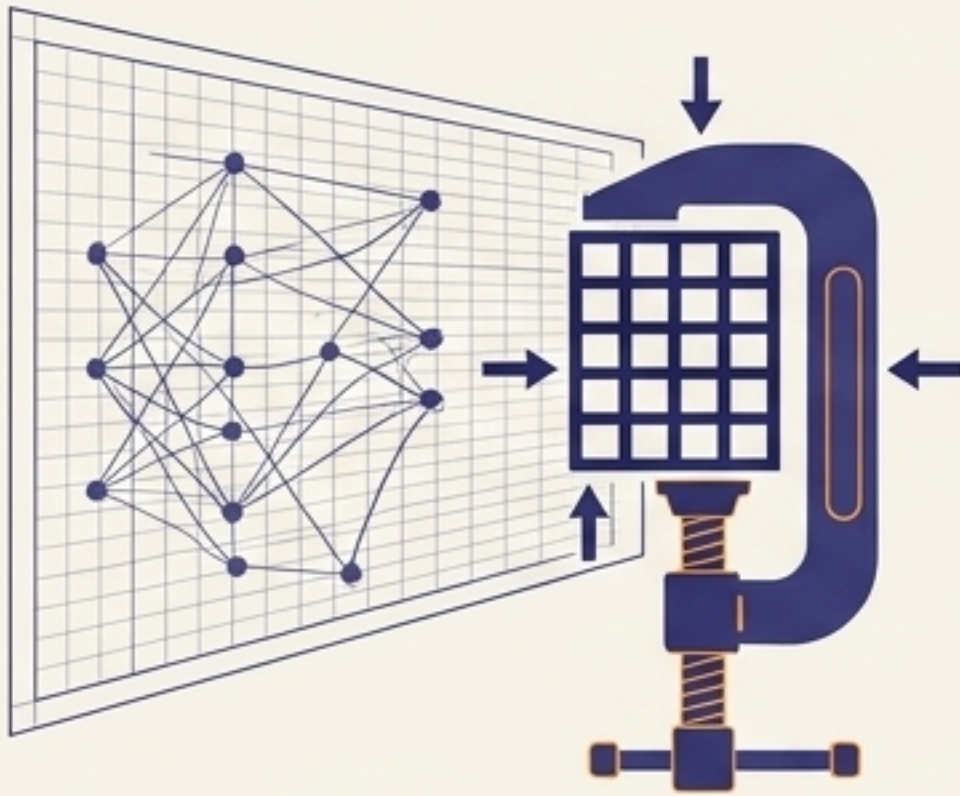


- Running AI models directly on local devices rather than remote servers:
 - **Privacy:** Sensitive user data never leaves the device.
 - **Offline Operation:** Independent predictions without internet connectivity.
 - **Bandwidth:** Only final insights are transmitted, reducing network strain.
 - **Constraints:** Engineers must overcome limited VRAM, compute power, and battery drain.

OPTIMISING MODELS FOR SPEED

Running billions of operations per second exceeds strict latency budgets. Software optimisation is required.

Quantisation



Trading a tiny fraction of mathematical precision for massive gains in processing speed and memory usage.

Pruning



Identifying and removing unnecessary 'weights' or connections to make the neural network lighter.

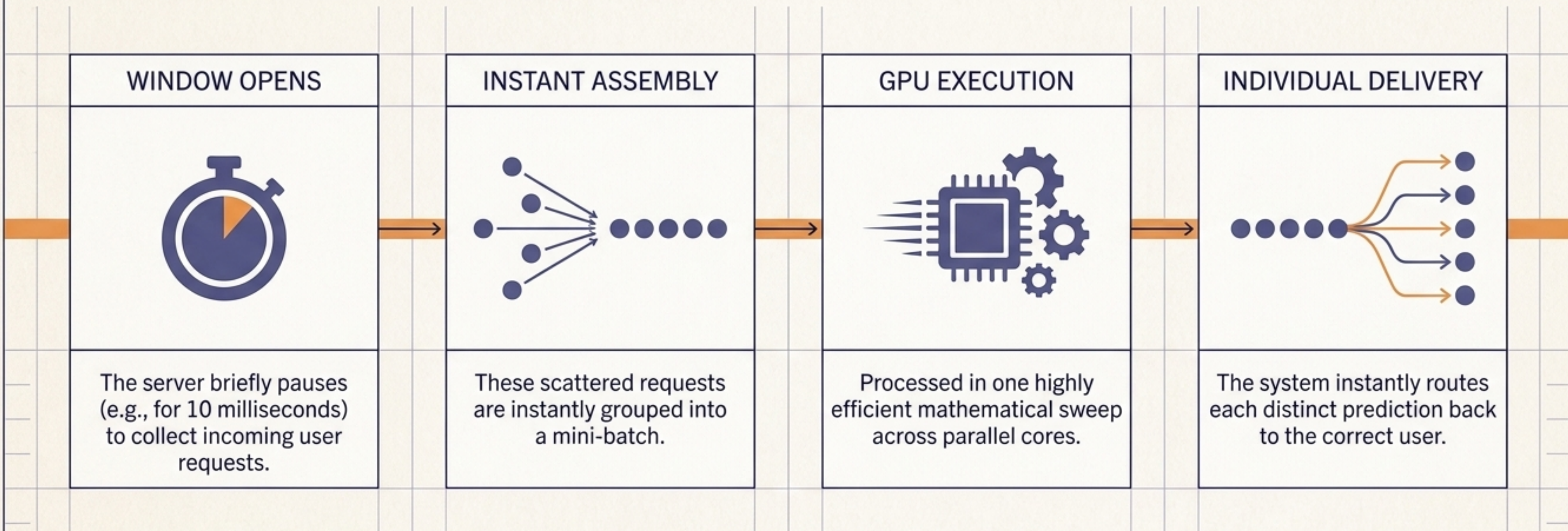
Knowledge Distillation



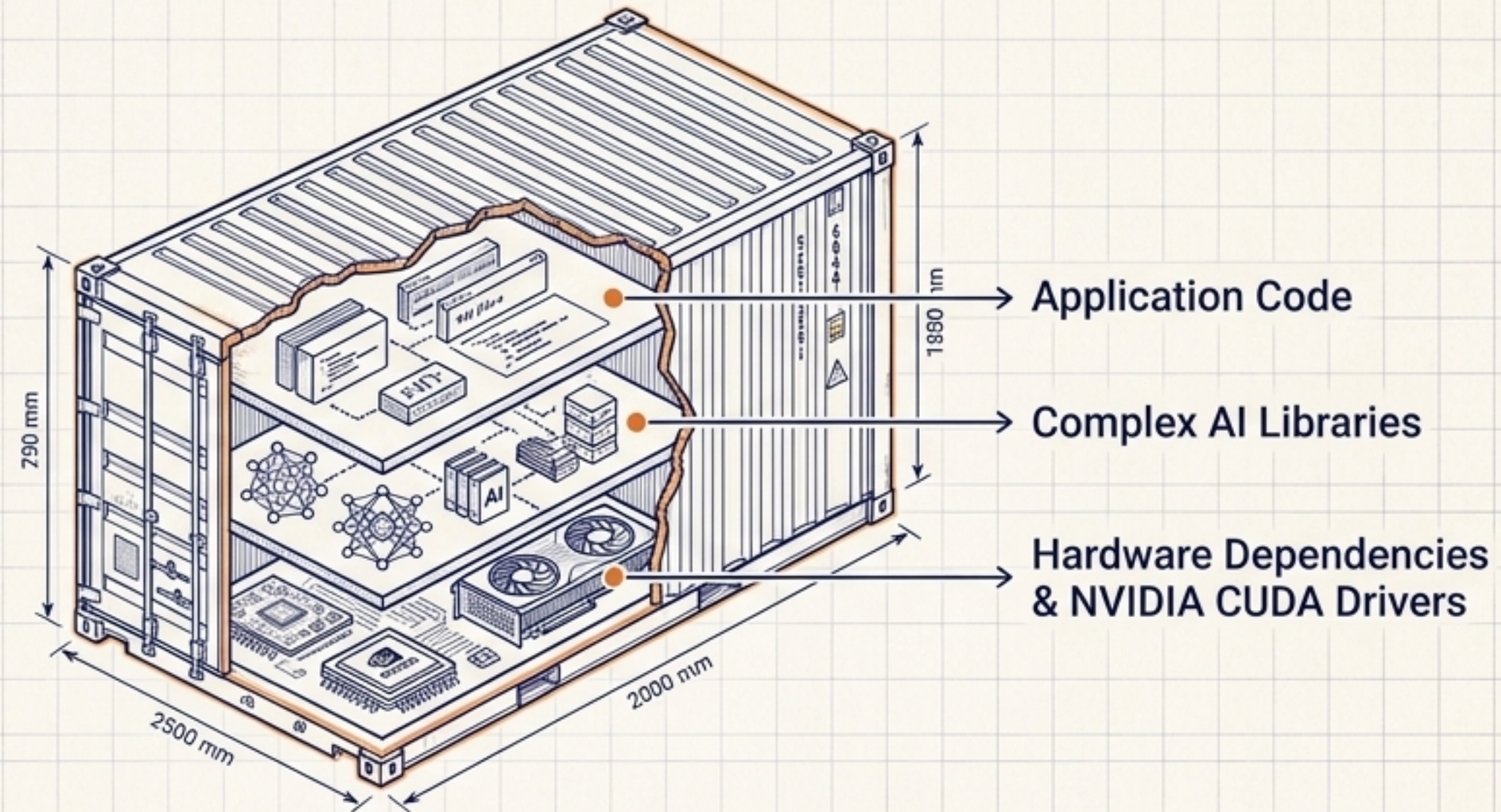
Using a massive, complex 'teacher' model to train a highly efficient 'student' model.

DYNAMIC BATCHING: THE BEST OF BOTH WORLDS

Combining the hardware efficiency of batching with real-time responsiveness.



CONTAINERISATION WITH DOCKER

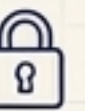


THE PROBLEM



AI environments are highly complex. Models frequently break when moved from a developer's laptop to a cloud server due to **mismatched dependencies**.

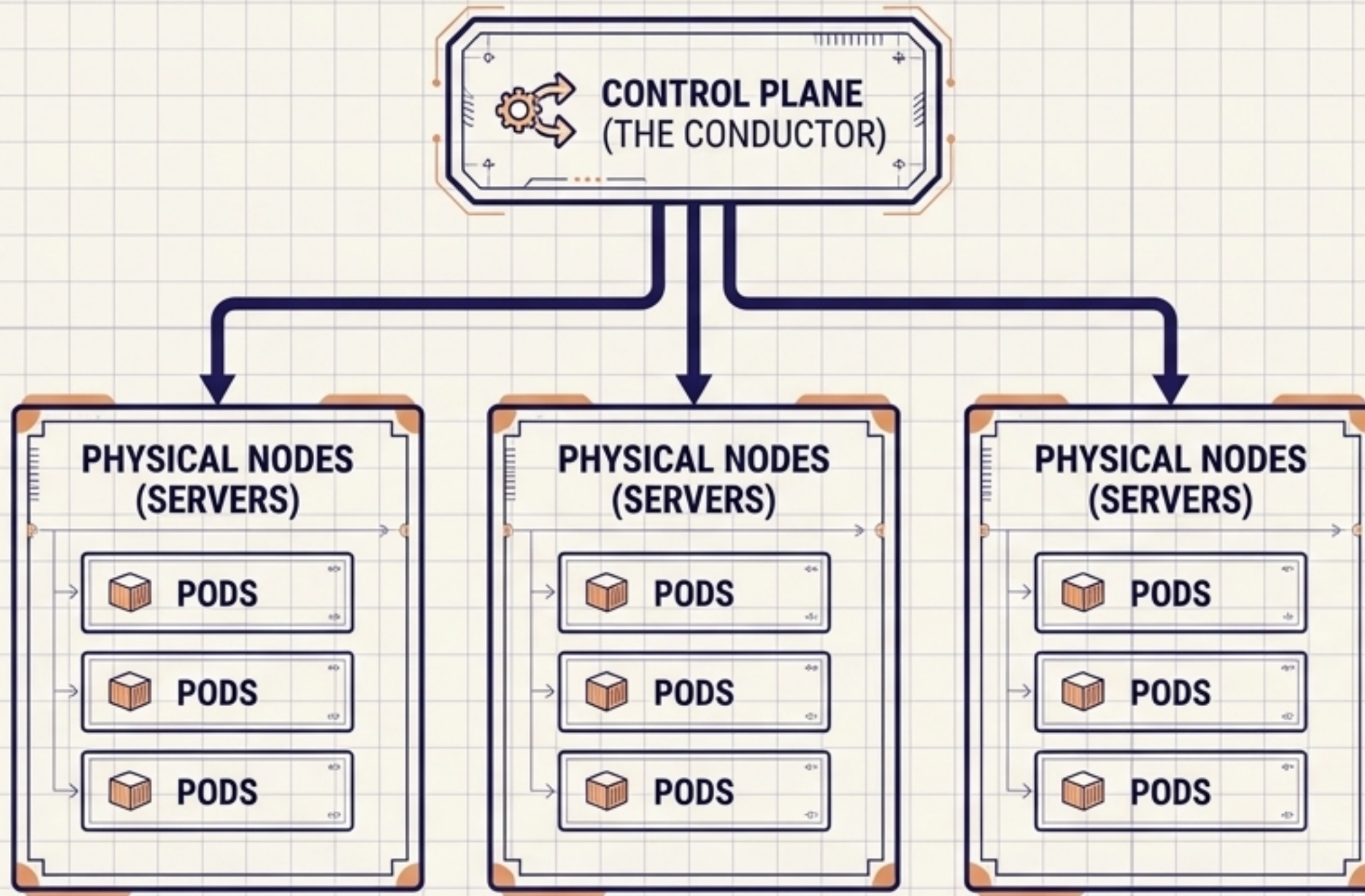
THE SOLUTION



Docker uses a simple text document (the **Dockerfile**) to package everything into one isolated, portable container. This ensures **100% reproducibility** across development, testing, and production.

ORCHESTRATION WITH KUBERNETES

Managing thousands of containers manually is impossible. **Kubernetes** automatically deploys, scales, and manages fleets.



MANAGING FLEETS AUTOMATICALLY

- **Pods and Nodes:** A 'Pod' is the smallest running unit containing your model, sitting on a physical 'Node'. 
- **Viral Growth:** Can automatically spin up hundreds of new Pods if an application experiences a massive spike in traffic. 
- **Self-Healing:** If an inference container crashes, Kubernetes detects the failure and instantly restarts it. 

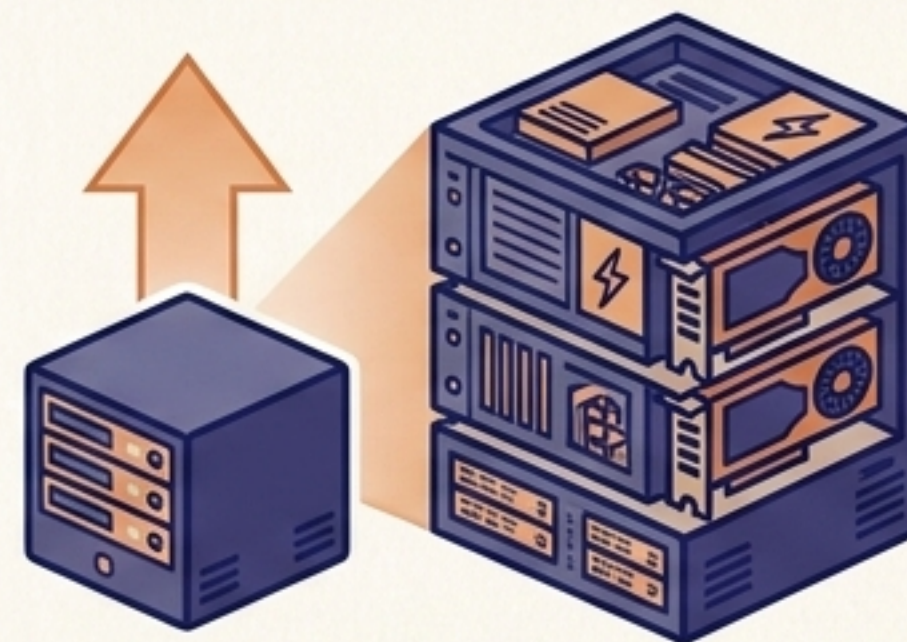
THE DYNAMICS OF SCALING HARDWARE

HORIZONTAL SCALING (SCALING OUT)



Adding more machines to handle a larger volume of concurrent users. Increases overall throughput but does not make a single request any faster.

VERTICAL SCALING (SCALING UP)



Upgrading a single machine to a larger, more powerful GPU to handle heavier models and reduce latency.

ADVANCED SCALING

Truly massive models require Tensor Parallelism—splitting the computation across multiple GPUs simultaneously. Asynchronous processing handles heavy inputs/outputs without freezing the entire system.

Observability: Monitoring AI in Production

Unlike traditional software that simply crashes, AI models can silently degrade and generate bad predictions.

PRODUCTION STABILITY

Metrics



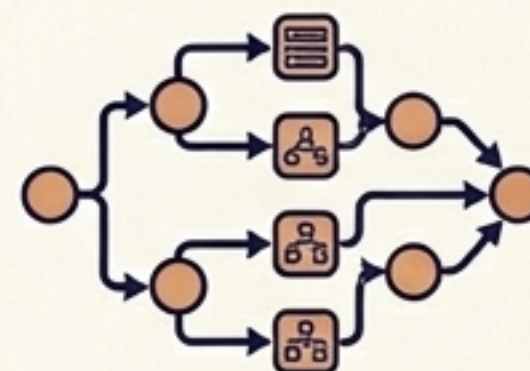
Tracking system vital signs like memory usage, latency spikes, and prediction volumes over time.

Logs



Capturing detailed text records of specific errors, system events, and application states.

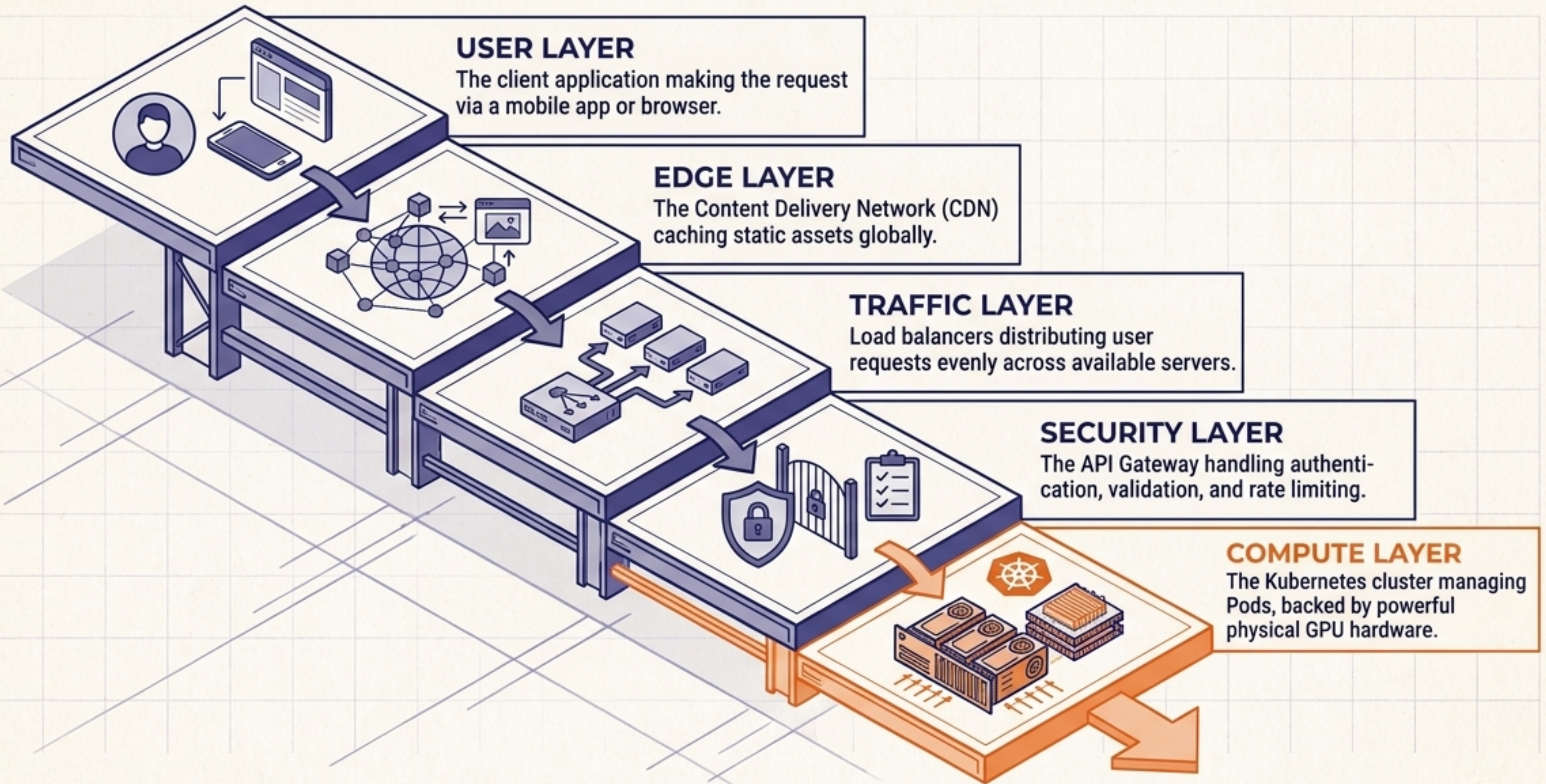
Distributed Traces



Following a single user request exactly as it travels through complex microservices.

Comprehensive observability must be active before the very first user request is served.

THE LAYERED PRODUCTION ARCHITECTURE



A robust, layered approach ensures scalability, security, and performance for high-demand AI services.

CORE PRINCIPLES FOR PRODUCTION AI ENGINEERING



01. VALUE CREATION

An AI model only creates real-world value once it is reliably served in production.



02. PARADIGM MATCHING

Choosing between batch, real-time, or edge dictates your entire infrastructure.



03. MULTI-LAYERED OPTIMISATION

True speed requires optimising both the model mathematics and the hardware.



04. NON-NEGOTIABLE TOOLS

Containerisation (Docker) and orchestration (Kubernetes) are mandatory for robust systems.



05. COST DISCIPLINE

GPU compute is highly expensive; engineering for efficiency is engineering for business survival.

• Adhering to these principles is critical for sustainable and scalable AI deployment.